SAÏFOULAYE DIALLO

Senior Backend Engineer

Java • Spring Boot • Distributed Systems • Banking & Payments

saifoulaye.diallo0512@gmail.com | +221 77 810 37 84 | Dakar, Senegal • Remote / International

linkedin.com/in/sayfoulaye-diallo | github.com/saifcores/saifcores | saifcore.tech | blog.saifcore.tech

PROFESSIONAL SUMMARY

Senior Backend Engineer with 4+ years owning production Java / Spring Boot systems — REST APIs, microservices and Kafka event flows for banking and payment platforms. Designs service boundaries, API contracts and reliability patterns (idempotency, transactional outbox, audit, reconciliation), then ships them to production (Docker, Kubernetes, CI/CD). Technical ownership with product, QA and engineering teams. Based in Dakar (GMT), bilingual French / English.

CORE EXPERTISE

Backend: Java, Spring Boot, Spring Security, Spring Data, REST APIs, Event-driven Architecture, OpenAPI, OAuth2, JWT

Architecture: Software Architecture, Distributed Systems, Microservices, Modular Monolith, API Design, System Integration, Legacy Modernization

Data & Messaging: PostgreSQL, MySQL, Oracle, MongoDB, Kafka, RabbitMQ, Redis, Flyway, Transactional Outbox

Reliability: Idempotency, audit logging, reconciliation, high availability, retries, financial correctness under partial failure

Cloud & Platform: Docker, Kubernetes, AWS, CI/CD, Terraform, Linux, GitLab CI, Jenkins

FinTech / Payments: Payment APIs, Mobile Money, Double-entry ledgers, Payment Orchestration, Transaction Processing, Webhook HMAC

PROFESSIONAL EXPERIENCE

Senior Backend Engineer — Banking Platforms (Consultant)

Feb. 2025 – Present

Synapse Groupe — Client: Bank of Africa Group — *Dakar, Senegal*

- Drove backend modernization of enterprise banking platforms (Java / Spring Boot, microservices, clean architecture) serving 11+ Bank of Africa Group subsidiaries — standardizing integration patterns and raising maintainability of core services.
- Designed and evolved critical backend services for payments, transfers and transaction processing on high-volume systems.
- Designed secured enterprise REST APIs and gateway-style boundaries connecting banking applications, middleware and third-party systems — versioned contracts, JWT / RBAC, audit logging, isolation of core systems.
- Implemented append-only ledger-style writes and transaction reconciliation so every money movement remains explainable for finance, audit and compliance.
- Industrialized operational resilience and high availability (Docker, Kubernetes, CI/CD) for critical production services.
- Held technical ownership with product, QA and multiple engineering teams — code reviews, software quality, Agile Scrum. *Environment:* Java, Spring Boot, Docker, Kubernetes, CI/CD, Microservices, Clean Architecture, Scrum.

Backend Engineer — FinTech & Payments

Jul. 2023 – Feb. 2025

ENG Technologie Group — *Dakar, Senegal*

- Owned backend microservices for digital credit, customer onboarding and payment workflows on a FinTech platform.
- Integrated Wave and Orange Money; designed Kafka event flows and explicit idempotency so retries cannot double-debit.
- Hardened APIs with Spring Security and improved payment operations reliability in production (AWS, Docker, GitLab CI/CD). *Environment:* Java, Spring Boot, AWS, PostgreSQL, Apache Kafka, Docker, GitLab CI/CD, Scrum.

Software Engineer — SaaS Platform

Jun. 2022 – Nov. 2023 (parallel ENG from Jul. 2023)

Mafalia — Ecom & Fintech Applications — *Dakar, Senegal*

- Owned backend services and REST APIs for a multi-tenant SaaS — inventory, orders, sales and financial workflows for SMEs.
- Improved performance through query optimization, caching and asynchronous processing, keeping reporting off the collection path.
- *Environment:* Java, Spring Boot, MySQL, REST APIs, Agile XP.

Software Engineer

Dec. 2021 – May 2022

SarayaTech — *Senegal*

- Delivered backend services and domain features for patient management, consultations and prescriptions on a clinical platform.
- Designed secured REST APIs across business modules; improved performance with caching and asynchronous processing.
- *Environment:* Java, Spring Boot, REST APIs, Scrum.

SELECTED PROJECTS

Ecom 360 PME Live SaaS • Senegal / West Africa

- Commerce platform: POS, real-time inventory, Wave & Orange Money on each sale, multi-store ops and accountant-ready exports — 10+ merchants live. ecom360pme.com
- Multi-tenant Spring Boot REST API, async reporting / export jobs and role-based staff access under network and wallet constraints.

Double-Entry Ledger Platform FinTech reference • GitHub

- Modular monolith (Java 21 / Spring Boot 3.4) for wallets and double-entry flows: balanced append-only journals, idempotent posting, transactional outbox to Kafka, maker/checker withdrawals.
- PostgreSQL + Flyway triggers, Redis rate limits, OpenAPI, Compose demo UI on the real API — posting invariants first, not a balances table.

Payment Platform Disaster Lab Resilience lab • GitHub

- Distributed XOF payment lab: dual PSP simulators, payment state machine, Kafka transactional outbox — prove no payment lost and none processed twice under partial failure.
- Timeout = UNKNOWN / IN_RECONCILIATION (never blind failover); SRE control center to inject chaos and observe recovery.

Pan-African Payment SDK Payment infrastructure

- Single API to initiate, confirm and reconcile Orange Money, Wave and Free Money — provider adapters behind one contract.
- Spring Boot adapters, Redis idempotency keys, Kafka settlement events, signed requests and HMAC webhook validation. Patterns proven on ENG Technologie FinTech delivery.

TECHNICAL STACK

Languages: Java (8–21), SQL, TypeScript

Backend: Java, Spring Boot, Node.js / NestJS, Spring Security, Spring Data JPA, Spring Cloud Gateway, REST APIs, OpenAPI / Swagger, OAuth2, JWT

Data & messaging: PostgreSQL, Oracle, MySQL, Redis, MongoDB, Kafka, RabbitMQ

Architecture: Microservices, modular monolith, clean / hexagonal architecture, DDD, Event-Driven, API-First, double-entry ledger patterns

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, GitHub Actions, GitLab CI, Jenkins, Maven, Git

Also: Angular, React, Next.js — when the API surface needs a client

EDUCATION

Master's, Software Engineering May 2024 – Present
Université numérique Cheikh Hamidou KANE (UN-CHK) *Dakar, Senegal*

Bachelor's, Mathematics & Computer Science Jan 2020 – May 2024
Université numérique Cheikh Hamidou KANE (UN-CHK) *Dakar, Senegal*

Computer Programming BootCamp Jan 2022 – Oct 2022
SarayaTech *Senegal*

LANGUAGES

French — Native / professional bilingual | English — Professional (business / technical writing)